

Li Lai

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Work Experience

- 2025–Present **Xiamen University**, School of Mathematical Sciences.
Associate Professor
- 2023–2025 **Peking University**, Beijing International Center for Mathematical Research.
Postdoc, Mentor: Liang Xiao
- 2020–2021 **Fudan University**, School of Mathematical Sciences.
Research Assistant, Mentor: Yijun Yao

Education

- 2021–2023 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
Ph.D. Mathematics, Advisor: Pin Yu
- 2014–2020 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
M.S. Mathematics, Advisor: Pin Yu
- 2010–2014 **Tsinghua University**, Department of Mathematical Sciences, Beijing, China.
B.S. Mathematics

Research Interests

I mainly work on transcendental number theory. I am interested especially in odd zeta values $\zeta(2k+1)$, p -adic odd zeta values $\zeta_p(2k+1)$ and multiple zeta values $\zeta(k_1, k_2, \dots, k_r)$.

Publications

7. Li Lai, Cezar Lupu and Derek Orr,
Elementary proofs of Zagier's formula for multiple zeta values and its odd variant,
Proc. Amer. Math. Soc. 154 (2026), no. 01, 11–24.
[arXiv:2201.09262](https://arxiv.org/abs/2201.09262)
6. Li Lai, Cezar Lupu and Johannes Sprang,
On the irrationality of certain p -adic zeta values,
Res. Math. Sci. 12 (2025), no. 4, Paper No. 77.
[arXiv:2505.23088](https://arxiv.org/abs/2505.23088)
5. Li Lai,
On the irrationality of certain 2-adic zeta values,
Int. J. Number Theory 21 (2025), no. 1, 207–235.
[arXiv:2304.00816](https://arxiv.org/abs/2304.00816)
4. Li Lai, Jiong-Yue Li and Pin Yu,
On the rigidity of stationary charged black holes: small perturbations of the non-extremal Kerr-Newman family,
J. Differential Geom. 125 (2023), no. 3, 553–612.
[arXiv:1911.10560](https://arxiv.org/abs/1911.10560)

3. Steven Charlton, Herbert Gangl, Li Lai, Ce Xu and Jianqiang Zhao,
On two conjectures of Sun concerning Apéry-like series,
Forum Math. 35 (2023), no. 6, 1533–1547.
arXiv:2210.14704
2. Li Lai and Li Zhou,
At least two of $\zeta(5), \zeta(7), \dots, \zeta(35)$ are irrational,
Publ. Math. Debrecen 101/3–4 (2022), 353–372.
arXiv:2103.00904
1. Li Lai and Pin Yu,
A note on the number of irrational odd zeta values,
Compos. Math. 156 (2020), no. 8, 1699–1717.
arXiv:1911.08458

Preprints

6. Li Lai and Jia Li,
A partial result towards the Chowla–Milnor conjecture,
arXiv:2505.12687
5. Li Lai, Johannes Sprang and Wadim Zudilin,
A note on the irrationality of $\zeta_2(5)$,
arXiv:2505.05005
4. Li Lai,
A note on the number of irrational odd zeta values, II,
arXiv:2501.05321
3. Li Lai,
Small improvements on the Ball–Rivoal theorem and its p -adic variant,
arXiv:2407.14236v2
2. Li Lai and Johannes Sprang,
Many p -adic odd zeta values are irrational,
to appear in Michigan Math. J.,
arXiv:2306.10393
1. Li Lai,
On the largest prime divisor of $n! + 1$,
to appear in Bull. Aust. Math. Soc.,
arXiv:2103.14894

Awards and Honors

- 2012 Nianzeng Sun Mathematical Analysis Award (at Tsinghua University)
- 2010 51st International Mathematical Olympiad: Gold Medal

Teaching

- Fall 2024 Rational Functions and the Irrationality of Odd Zeta Values, Mini Course, Peking University
- Spring 2024 Advanced Mathematics B (2), Peking University

Spring 2021 Rational Functions and the Irrationality of Odd Zeta Values, Short Course, Fudan University

Seminar (Co)Organized

Fall 2021–Spring 2022 Tsinghua-BIMSA Learning Seminar on Multiple Zeta Values, Tsinghua University

Other Experiences and Activities

2017 Finisher of Columbia168 ULTRA-TRAIL® THREE GORGES (168 km trail running)

Spring 2013 Exchange student at École Normale Supérieure, Paris, France

CV updated: 2026-1-5